ASSIGNMENT:- 5

AIM:- Data Visualization

PROBLEM STATEMENT:- Connect to data, Build Charts and Analyze Data, Create Dashboard, Create Stories using Tableau/PowerBI.

THOERY:-

Data Visualization is a critical step in the data analysis process, allowing users to interpret data through interactive and visual representations. Tools like Power BI and Tableau are widely used in industry for connecting to various data sources, building meaningful visualizations, creating dashboards, and telling interactive data stories that drive business insights and decision-making.

This assignment involves working with a real-world dataset titled *"Data Professionals Survey"*, and demonstrating the complete workflow in both Power BI or Tableau — from data connection and transformation to interactive dashboards and story-building.

Objectives

- Connect to and clean data using Tableau or Power BI.
- Create various types of charts and visualizations.
- Analyze trends and derive insights.
- Design dashboards to present KPIs and metrics.
- Use storytelling features to guide end-users through insights interactively.

Dataset Description

The dataset data_professionals_survey.csv contains responses from data professionals around the world, covering aspects such as:

- Age and Salary
- Job Titles
- Country of residence
- Preferred programming languages
- Happiness with salary and work
- Perceived difficulty of entering the data profession

This dataset helps us understand demographics, salary distribution, tool preferences, and satisfaction levels among data professionals.

Implementation Steps

1. **Connect to Data**
   - **Power BI:-** Used “Get Data” → CSV → Loaded the dataset and transformed data using Power Query Editor for cleaning (e.g., renaming columns, handling null values).
   - **Tableau:-** Used “Text File” connection to load the dataset. Pivoted programming language columns for unified analysis and checked data types for consistency.

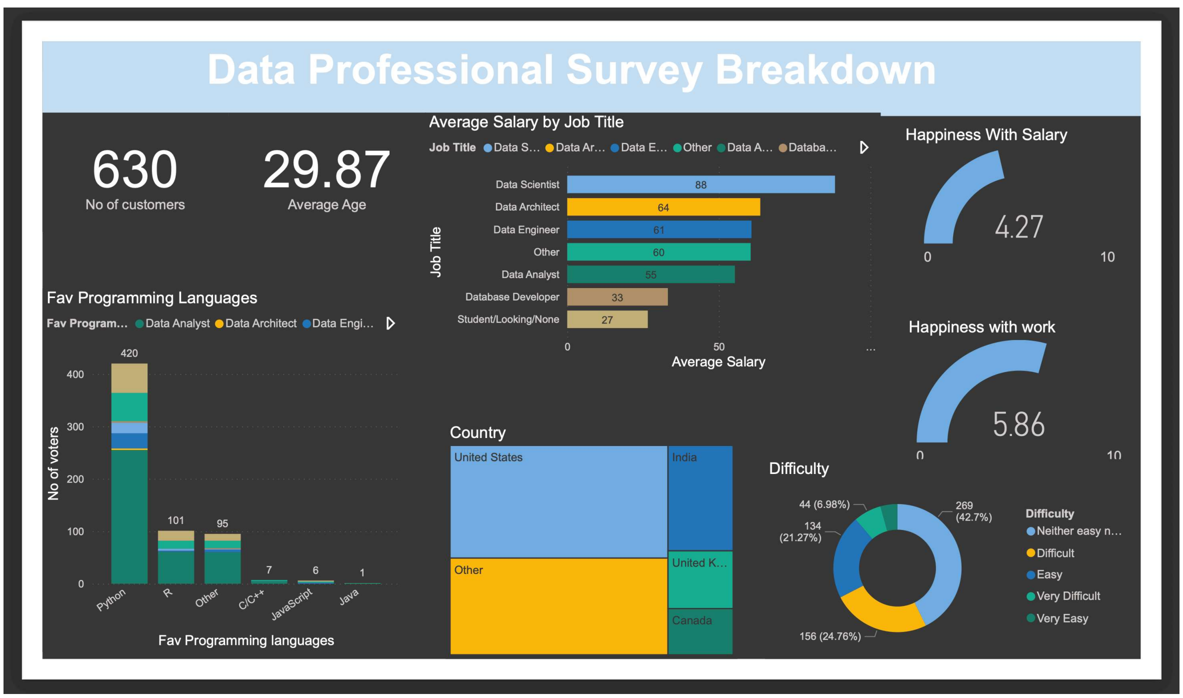
2. Build Charts and Analyze Data
Different charts were created to explore the dataset:

- **Bar Charts:-** To analyze job title distribution and most preferred programming languages.
- **Line Charts:-** To visualize trends like salary across age groups.
- **Pie/Donut Charts:-** To show the country-wise distribution.
- **KPI Cards:-** To display average age, total respondents, and average salary.
- **Table/Matrix Views:-** For detailed job title and salary breakdown.
- **Maps (in Tableau):-** For geographic distribution of participants.

DAX formulas in Power BI and Calculated Fields in Tableau were used to derive additional insights like high-earning roles, average happiness scores, etc.

3. Create Dashboards
Dashboards were built by logically arranging the visuals with consistent themes:

- Overview Dashboard: KPIs, job title chart, and country distribution.
- Salary Insights: Salary by job title and satisfaction scores.
- Technical Preferences: Programming languages and tool choices.
- Filters like Country, Job Title, and Age were added for interactivity.



CONCLUSION:-
This assignment demonstrated how Tableau and Power BI can be used to uncover valuable insights from a dataset through visual storytelling. It highlighted the importance of effective data cleaning, appropriate visualization selection, interactive design, and storytelling in deriving and communicating insights. Both tools empowered us to explore data in depth and share findings in a meaningful and visually engaging way.